

ENTERTAINMENT OS

Automation Framework

Google Cloud Platform • Vertex AI • n8n Orchestration

"Calm during chaos. Making the complex simple."

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Executive Summary

This document defines the comprehensive automation framework for building the Entertainment OS using Google Cloud Platform (GCP), Vertex AI for machine learning and predictive intelligence, and n8n for workflow orchestration. The framework transforms the existing EVERGAME architecture documentation into an automated, continuously deployable platform capable of operating at enterprise scale.

Strategic Objective

Automate the complete Entertainment OS buildout using Infrastructure-as-Code (Terraform), containerized deployments (GKE), AI-powered predictions (Vertex AI), and visual workflow orchestration (n8n) to achieve sub-10ms response times, 94%+ prediction accuracy, and 99.97% platform availability.

Document Scope

This framework covers seven interconnected automation domains that together enable full-stack Entertainment OS deployment:

Domain	Technology Stack	Automation Target
Infrastructure Provisioning	Terraform, GCP, GKE	Zero-touch environment creation
ML Model Pipeline	Vertex AI, BigQuery, Dataflow	Automated training & deployment
Workflow Orchestration	n8n, Redis, PostgreSQL	Visual workflow management
Data Simulation	Synthetic Framework, TimescaleDB	Demo-ready venue telemetry
Evidence Capture	Blockchain-style ledger, GCS	Immutable audit trails
Prediction Engine	Vertex AI Endpoints, Cloud Run	Real-time inference at scale
Continuous Deployment	GitHub Actions, Cloud Build	Automated testing & release

1. Google Cloud Platform Architecture

The Entertainment OS leverages GCP's enterprise-grade infrastructure to deliver high-availability, low-latency operations for live event orchestration. The architecture follows a multi-region, defense-in-depth approach with complete Infrastructure-as-Code automation.

1.1 Core Infrastructure Components

Component	GCP Service	Configuration	Purpose
Compute Cluster	Google Kubernetes Engine	Regional, 3 zones, auto-scaling	Container orchestration for all services
Primary Database	Cloud SQL (PostgreSQL 15)	HA, 4 vCPU, 16GB RAM	Relational data: events, staff, incidents
Document Store	MongoDB Atlas (GCP)	M30 cluster, 3 nodes	Flexible schemas: SOPs, blueprints
Time-Series DB	TimescaleDB on GKE	Compressed hypertables	Sensor telemetry, crowd metrics
Caching Layer	Memorystore (Redis)	4GB, HA replicas	Session state, real-time dashboards
Object Storage	Cloud Storage	Multi-regional, lifecycle	Evidence archives, ML artifacts
API Gateway	Kong on GKE	Rate limiting, auth	Unified API entry point
ML Platform	Vertex AI	Custom containers	Prediction models & endpoints

1.2 Terraform Module Structure

Infrastructure provisioning follows a modular Terraform approach with environment-specific configurations:

Directory Structure

terraform/ ├── environments/ | ├── dev/ | ├── staging/ | └── production/ ├── modules/ | ├── gke-cluster/ | ├── cloud-sql/ | ├── vertex-ai/ | ├── networking/ | └── monitoring/ └── shared/

1.3 Network Architecture

Network Layer	CIDR Range	Components
VPC Primary	10.0.0.0/20	GKE nodes, internal services
Pod Network	10.4.0.0/14	Kubernetes pod IPs
Service Network	10.8.0.0/20	Kubernetes service IPs
Private Services	10.16.0.0/20	Cloud SQL, Redis, MongoDB

2. Vertex AI Machine Learning Pipeline

The prediction engine leverages Google Vertex AI for end-to-end ML operations, from feature engineering through model deployment and monitoring. The pipeline automates training, evaluation, and deployment of the five core prediction models that power EVERGAME intelligence.

2.1 Prediction Models Overview

Model	Architecture	Accuracy	Latency	Use Case
Crowd Behavior	LSTM Neural Network	94%	<50ms	Density prediction, flow optimization
Equipment Failure	Random Forest Ensemble	92%	<100ms	30-day predictive maintenance
Weather Impact	Ensemble (GBM + LSTM)	96%	<200ms	Operational adjustment triggers
Revenue Optimization	Reinforcement Learning	12% lift	<150ms	Dynamic pricing recommendations
Security Threat	Anomaly Detection (Isolation Forest)	0.001% FP	<30ms	Real-time threat identification

2.2 Vertex AI Pipeline Components

Each model follows a standardized pipeline architecture with automated stages:

Pipeline Stage	GCP Service	Automation	Output
Data Ingestion	BigQuery, Cloud Storage	Scheduled Dataflow jobs	Feature tables, training datasets
Feature Engineering	Vertex Feature Store	Automated feature pipelines	Versioned feature sets
Model Training	Vertex AI Training	Hyperparameter tuning jobs	Trained model artifacts
Model Evaluation	Vertex AI Experiments	Automated metric comparison	Champion/challenger analysis
Model Registry	Vertex Model Registry	Version control, lineage	Production-ready models

Endpoint Deployment	Vertex AI Endpoints	Canary deployments	Serving infrastructure
Monitoring	Vertex Model Monitoring	Drift detection, alerts	Retraining triggers

2.3 Feature Store Architecture

Vertex Feature Store provides centralized feature management with online (low-latency) and offline (batch) serving capabilities:

Feature Groups

crowd_features: zone_density, flow_rate, dwell_time, entry_velocity (15-sec refresh)
equipment_features: vibration_pattern, temperature_delta, power_consumption, runtime_hours
weather_features: temperature, humidity, wind_speed, lightning_probability, precipitation
transaction_features: concession_sales, merchandise_velocity, parking_utilization
security_features: anomaly_score, crowd_sentiment, access_patterns, incident_history

3. n8n Workflow Orchestration

n8n serves as the visual workflow orchestration layer, connecting all Entertainment OS components into executable operational intelligence. The architecture follows the EVERGAME four-zone lifecycle (PREPARE → READY → RUN → REVIEW) with automated state transitions and evidence capture.

3.1 Workflow Categories

Category	Workflow Count	Purpose	Trigger Type
LIFECYCLE_*	4 master workflows	Zone phase orchestration	State machine transitions
MONITOR_*	8 polling loops	Real-time system observation	Cron (30-second intervals)
RESPONSE_*	12 incident handlers	Emergency response automation	Webhook + manual
PREDICTION_*	5 ML integrations	Vertex AI model invocations	Scheduled + event-driven
INTEGRATION_*	15 connectors	External system synchronization	API polling + webhooks
REPORT_*	6 generators	Automated documentation	Scheduled + on-demand
SHARED_*	10 utilities	Common functions reuse	Sub-workflow calls

3.2 n8n Infrastructure Configuration

Configuration	Local/Dev	Staging	Production
Execution Mode	regular	queue	queue
Worker Instances	0	4	4-8 (auto-scale)
PostgreSQL Connections	20	100	200
Redis Memory	256MB	2GB	4GB
API Rate Limit (per min)	1000	200	100
WebSocket Connections	100	2000	10,000
Dashboard Refresh	30s	30s	30s

3.3 Vertex AI Integration Workflows

n8n workflows invoke Vertex AI endpoints for real-time predictions with automatic fallback and caching:

Workflow	Model Endpoint	Cache TTL	Fallback Strategy
PREDICTION_Crowd_Density	crowd-behavior-v3	15 seconds	Last known good + exponential smoothing
PREDICTION_Equipment_Health	equipment-failure-v2	5 minutes	Conservative thresholds, alert ops
PREDICTION_Weather_Impact	weather-impact-v1	10 minutes	External weather API backup
PREDICTION_Revenue_Optimize	revenue-rl-v1	30 seconds	Static pricing rules
PREDICTION_Threat_Detect	security-anomaly-v1	No cache	Manual security escalation

4. Synthetic Data & Simulation Automation

The Synthetic Data Framework enables full-fidelity Entertainment OS operation without live venue connections. Automation covers venue profile generation, sensor telemetry simulation, incident scenario injection, and temporal replay at configurable speeds.

4.1 Generation Modules

Module	Output Type	Automation Capability
Venue Profile Generator	PostgreSQL + MongoDB records	Complete venue with zones, sensors, staff, SOPs
Sensor Telemetry Generator	TimescaleDB time-series	Realistic patterns with anomaly injection
Crowd Simulation Engine	Flow metrics, density maps	Agent-based modeling with event patterns
Weather Scenario Generator	Weather API mock responses	Clear day, approaching storm, lightning strike
Incident Scenario Library	Structured incident objects	8 SOP templates with compliance mapping
Evidence Ledger Seeder	Hash-chained records	Verifiable audit trail generation
Temporal Simulation Engine	Real-time or accelerated replay	1x to 300x speed with n8n webhook triggers

4.2 Demo Scenario Packages

Pre-composed scenario bundles designed for specific audiences with configurable playback speeds:

Package	Duration @ 60x	Audience	Key Demonstrations
Perfect Game Day	5 minutes	Executive, Investor	Dashboard visualization, guest experience metrics
Lightning Strikes	8 minutes	Operations, Safety	SOP execution, evidence capture, all-clear
Cascading Crisis	12 minutes	Federal, Compliance	Multi-incident coordination, NIMS protocols

Full Season	20 minutes	Technical, Engineering	Complete workflow coverage, edge cases
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Temporal Engine Control API

POST /api/simulate/start - Begin scenario playback
POST /api/simulate/pause - Pause at current simulation time
POST /api/simulate/resume - Continue from paused state
POST /api/simulate/speed - Adjust playback speed (1x-300x)
GET /api/simulate/status - Current state, time, events emitted
POST /api/simulate/inject - Manual incident injection

5. Continuous Integration & Deployment

The Entertainment OS follows a GitOps deployment model with automated testing, security scanning, and progressive rollouts. GitHub Actions orchestrates the pipeline with Cloud Build for container images and ArgoCD for Kubernetes deployments.

5.1 Pipeline Stages

Stage	Trigger	Actions	Success Criteria
Lint & Validate	PR opened	ESLint, Terraform validate, YAML lint	Zero errors
Unit Tests	PR opened	Jest, pytest, workflow tests	95%+ coverage
Security Scan	PR opened	Snyk, Trivy, SAST	No critical vulnerabilities
Build Images	PR merged to main	Cloud Build, multi-arch	Image push to Artifact Registry
Deploy Staging	Image tagged	ArgoCD sync, migrations	Health checks pass
Integration Tests	Staging deployed	E2E scenarios, load tests	All scenarios green
Deploy Production	Tag created (v*)	Canary rollout, monitoring	Error rate < 0.1%

5.2 Workflow Deployment Automation

n8n workflows are version-controlled and deployed through the CI/CD pipeline:

Workflow Deployment Process

1. Export workflows to JSON via n8n CLI 2. Commit to git repository (n8n/workflows/) 3. PR review with diff visualization 4. Automated validation against schema 5. Import to staging n8n instance 6. Execute test scenarios 7. Production import via kubectl exec

5.3 ML Model Deployment Pipeline

Stage	Automation	Output
Training Trigger	Cloud Scheduler (weekly) or drift alert	Vertex AI Training job

Hyperparameter Tuning	Vertex AI Vizier, 50 trials	Optimal configuration
Model Evaluation	Comparison against champion	Metrics dashboard update
Model Registration	Automatic if metrics improve	New model version
Endpoint Update	Canary deployment (10% → 50% → 100%)	Production serving
Monitoring Setup	Automatic drift detection config	Alert channels configured

6. Automation Scripts & Commands

Complete automation is achieved through a set of documented scripts and CLI commands that handle all operational tasks from environment provisioning to model deployment.

6.1 Infrastructure Commands

Command	Purpose
terraform init -backend-config=env/prod.tfbackend	Initialize Terraform with production backend
terraform plan -var-file=env/prod.tfvars -out=plan.out	Generate execution plan for review
terraform apply plan.out	Apply infrastructure changes
gcloud container clusters get-credentials sentrais-prod	Configure kubectl for production cluster
kubectl apply -k kubernetes/overlays/production/	Deploy Kubernetes resources via Kustomize
helm upgrade --install n8n charts/n8n -f values-prod.yaml	Deploy n8n with Helm

6.2 ML Pipeline Commands

Command	Purpose
gcloud ai custom-jobs create --config=training/crowd-model.yaml	Submit training job to Vertex AI
gcloud ai models upload --artifact-uri=gs://... --model-id=crowd-v3	Register model in Model Registry
gcloud ai endpoints deploy-model ENDPOINT_ID --model=MODEL_ID	Deploy model to serving endpoint
gcloud ai endpoints predict ENDPOINT_ID --json-request=input.json	Test prediction endpoint
vertex-ai-pipeline run --config=pipelines/full-retrain.yaml	Execute complete ML pipeline

6.3 Synthetic Data Commands

Command	Purpose
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<code>python -m sentrais_synth.cli venue seed --all</code>	Seed all venue profiles to databases
<code>python -m sentrais_synth.cli simulate run --scenario lightning_strikes</code>	Run simulation scenario
<code>python -m sentrais_synth.cli validate all --venue-id venue_mbs_001</code>	Validate data integrity
<code>docker run sentrais-synthetic uvicorn sentrais_synth.temporal.api:app</code>	Start simulation REST API

7. Monitoring & Observability Automation

Comprehensive monitoring ensures system health, performance targets, and SLA compliance. Automation covers metric collection, alerting, and incident response triggering.

7.1 Performance Targets

Metric	Target	Alert Threshold
API Response Time (p50)	< 50ms	> 100ms for 5 minutes
API Response Time (p99)	< 200ms	> 500ms for 3 minutes
Workflow Execution (monitoring)	< 5s per cycle	> 10s for 3 cycles
Dashboard Update	< 3s	> 5s for 5 updates
Prediction Latency	< 100ms	> 200ms for 1 minute
Platform Availability	99.97%	Any downtime > 30s
Workflow Failure Rate	< 5%	> 5% in 5 minutes

7.2 Alerting Configuration

Priority	Channel	Response SLA	Example Triggers
P1 - Critical	PagerDuty + Slack #incidents	< 5 minutes	Platform down, data corruption
P2 - High	PagerDuty + Slack #alerts	< 15 minutes	Workflow failures > 5%, API errors > 1%
P3 - Medium	Slack #ops-notifications	< 2 hours	Dashboard latency, non-critical failures
P4 - Low	Email digest	Next business day	Warning thresholds, optimization opportunities

7.3 Cloud Monitoring Setup

Automated Monitoring Resources

Cloud Monitoring Workspaces: sentrais-prod, sentrais-staging Custom Dashboards: Platform

Overview, ML Model Health, Workflow Execution Uptime Checks: API Gateway (30s), WebSocket Gateway (30s), n8n Health (60s) Log-based Metrics: Workflow duration, prediction latency, error patterns Alert Policies: 47 policies across 7 severity levels Notification Channels: PagerDuty, Slack, Email, SMS

8. Implementation Roadmap

The Entertainment OS automation buildout follows a phased approach, ensuring each layer is stable before building the next. The roadmap assumes a dedicated engineering team with GCP and ML expertise.

8.1 Phase Timeline

Phase	Duration	Deliverables	Success Criteria
1: Infrastructure Foundation	Weeks 1-3	Terraform modules, GKE cluster, databases	terraform apply succeeds, kubectl connectivity
2: n8n Orchestration	Weeks 4-6	n8n deployment, 20 core workflows	All LIFECYCLE workflows executing
3: Synthetic Data	Weeks 7-8	Data generators, 3 demo scenarios	Full simulation playback working
4: Vertex AI Integration	Weeks 9-12	ML pipelines, 5 model endpoints	All models serving with <100ms latency
5: Evidence & Compliance	Weeks 13-14	Evidence Ledger, hash verification	Audit trail integrity verified
6: Production Hardening	Weeks 15-16	Load testing, security audit, DR	99.97% availability demonstrated

8.2 Resource Requirements

Role	Allocation	Responsibilities
Senior Cloud Architect	Full-time	GCP infrastructure, Terraform, networking
ML Engineer	Full-time	Vertex AI pipelines, model training, feature engineering
Backend Developer	Full-time	n8n workflows, API integrations, database design
DevOps Engineer	Full-time	CI/CD, monitoring, security, deployment automation
QA Engineer	50%	Test automation, scenario validation, load testing

8.3 Estimated GCP Costs

Resource Category	Monthly Cost (Production)	Notes
GKE Cluster (n2-standard-4 x 6)	\$1,800	Auto-scaling 3-8 nodes
Cloud SQL (PostgreSQL HA)	\$650	4 vCPU, 16GB RAM, HA
Memorystore (Redis)	\$180	4GB HA configuration
Vertex AI Training	\$500-2,000	Variable based on training frequency
Vertex AI Prediction	\$800	Online prediction endpoints
Cloud Storage	\$150	Evidence archives, ML artifacts
Networking & Egress	\$300	API traffic, inter-region
Monitoring & Logging	\$200	Cloud Monitoring, custom metrics
TOTAL ESTIMATED	\$4,580-6,080/month	Production workload

Appendix A: Configuration Reference

Environment Variables

Variable	Description	Example
SENTRAIS_ENV	Deployment environment	production
GCP_PROJECT_ID	Google Cloud project	sentrais-prod-12345
N8N_ENCRYPTION_KEY	Workflow encryption key	(32-byte hex)
POSTGRES_DSN	PostgreSQL connection string	postgresql://...
MONGODB_URI	MongoDB Atlas connection	mongodb+srv://...
REDIS_URL	Redis connection string	redis://...
VERTEX_ENDPOINT_CROWD	Crowd model endpoint ID	123456789
VERTEX_ENDPOINT_WEATHER	Weather model endpoint ID	987654321
EVIDENCE_SIGNING_KEY	Evidence ledger signing key	(64-byte hex)

Appendix B: API Endpoint Reference

Endpoint	Method	Purpose
/api/v1/events	GET, POST	Event lifecycle management
/api/v1/incidents	GET, POST, PUT	Incident CRUD operations
/api/v1/crowd/:zone_id	GET	Real-time crowd density
/api/v1/predict/crowd	POST	Crowd prediction request
/api/v1/predict/equipment	POST	Equipment failure prediction
/api/v1/evidence/:event_id	GET	Evidence ledger retrieval
/api/v1/sops/:sop_id/execute	POST	SOP execution trigger
/ws/dashboard	WebSocket	Real-time dashboard updates
/health	GET	Platform health check

Document End

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sentrais_os_deployment_guide.md